Contents

Intro

Problems

Solution

Usage

Q&A

ODSC AI WEST 2025

Intro

Your Code is cheap

The Execution is your moat;

Design makes it possible.

Data Model defines how your company ships their products.

ODSC AI WEST 2025

What’s the problem?

- **Today’s Enterprises maintain the same business entities (ex: Customer, Transactions) — in dozens of disconnected schemas across different systems.**
  - ***APIs uses one definition,***
  - ***Kafka uses another,***
  - ***PostgreSQL use a third.***
  - ***And more…***
- **This fragmentation creates massive technical debt, slows down velocity of the company delivery and makes governance nearly impossible.**

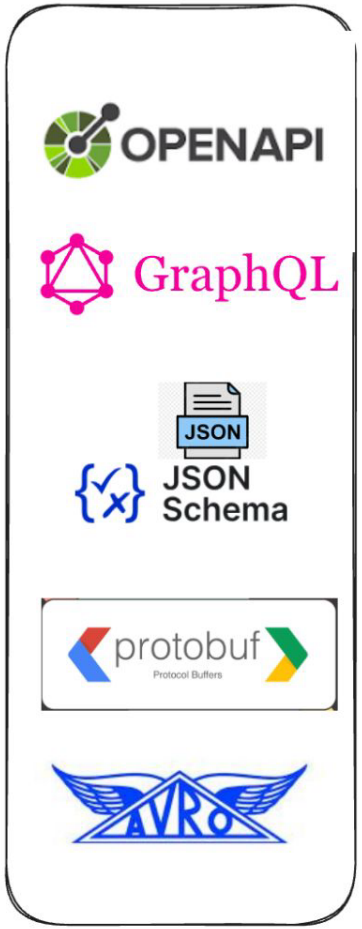
ODSC AI WEST 2025

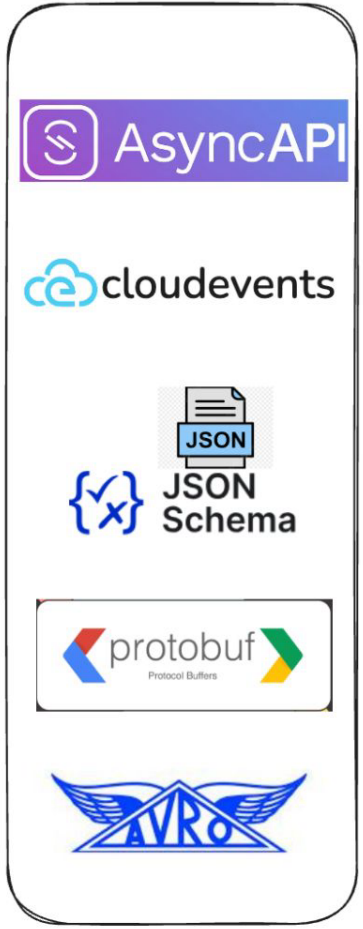
API

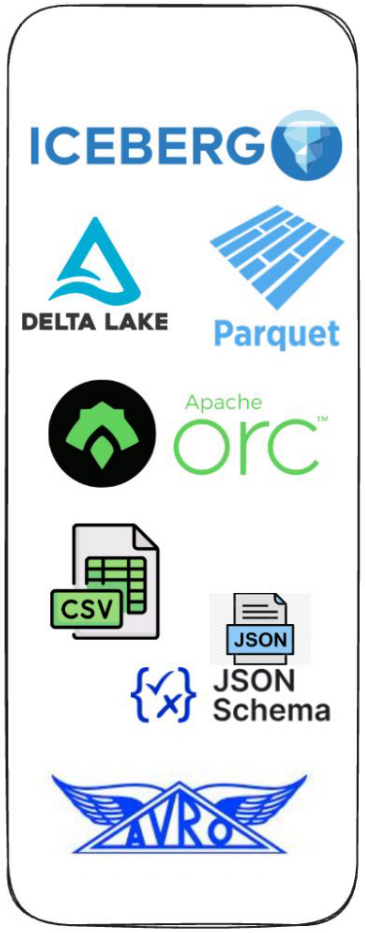
Events

BigData

Interface Definition Language







Governance

Metadata

Entity Model

Data Types

Why it’s a problem?

Technical chaos:

1. A customer's email is VARCHAR(255) in one system, VARCHAR(200) in another, and just 'string' in a third.
   - Every schema change requires coordinating 5+ teams - reduces velocity, increases risk.
   - Inconsistencies causes lot of confusions & data quality issues - Indirect loss of productivity or revenue.
2. Product Price in-consistency

PostgreSQL → NUMERIC(10,2) = 19.99 ✓ (exact)
Kafka/Avro → double = 19.989999... ✗ (precision loss)
Service API → float = 19.98 ✗ (worse!)
Customer Sees → $19.98 = Lost $0.01

For a billion transactions - 10 Million revenue loss or variation in analytics insights.

This directly impacts risk & velocity of the organization

ODSC AI WEST 2025

Why it’s a problem?

Governance nightmare: – Loss of productivity & increased compliance risk.

- Difficult to catalog; ‘'where is customer data stored?’ -
- Data classification & Retention policies lives in spreadsheets and its tribal knowledge
- Data protection regulations are hard to maintain across dozens of teams & systems.

Frontend Team → TypeScript types (400 lines)
API Team → OpenAPI Schema (500 lines)
Backend Team → PostgreSQL DDL (450 lines)
Mobile Team → Protobuf (350 lines)
Data Team → Avro Schema (600 lines)
Analytics Team → BigQuery DDL (550 lines)
ML Team → Parquet Schema (300 lines)
Governance Team → Excel Spreadsheet (200 rows)
Documentation → Confluence Page (outdated)

ODSC AI WEST 2025

Proposing a – “Super Schema”

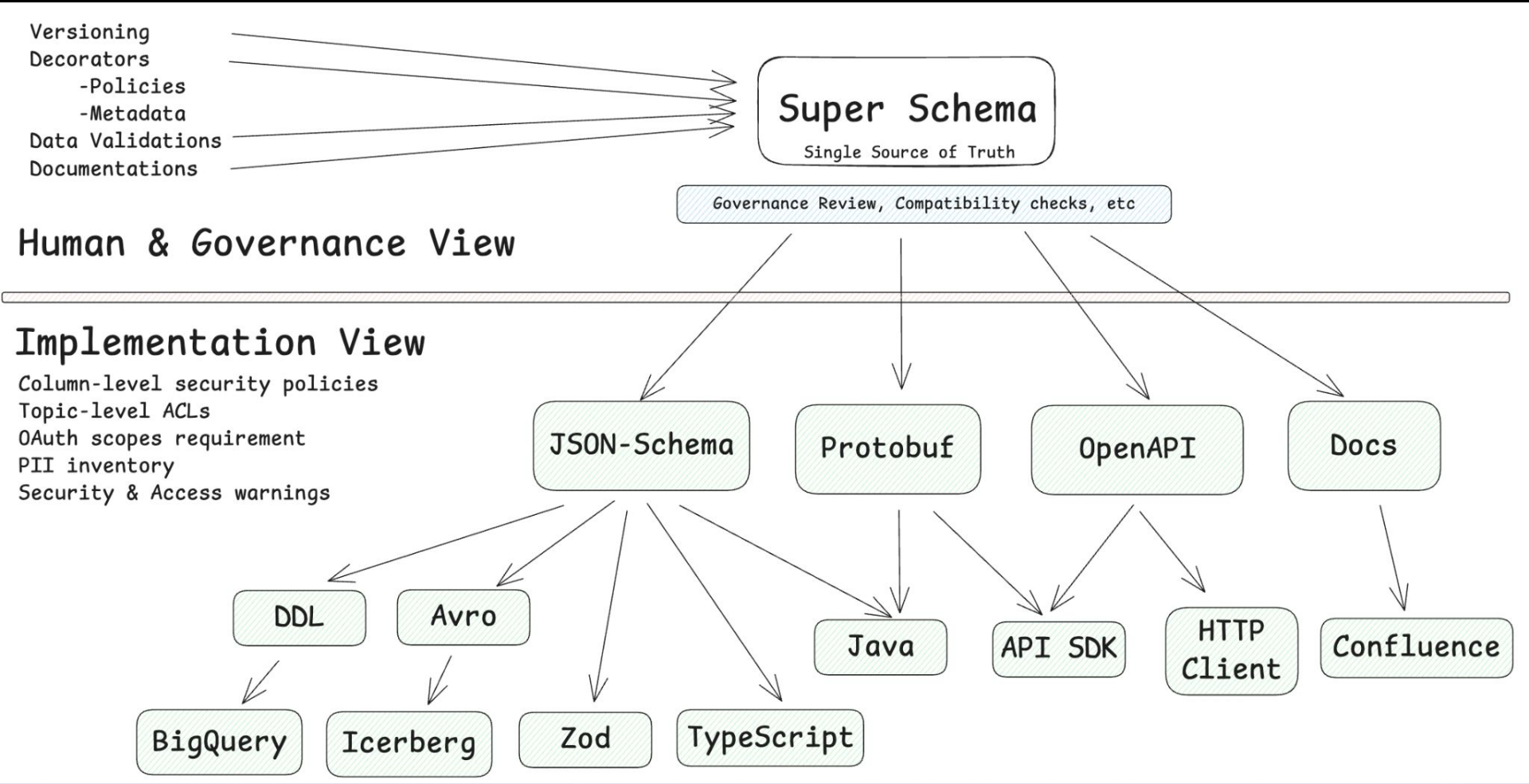
A versioned, human-readable logical schema expressed in non-technically-attached syntax that defines entities in an organization’s domain vocabulary, relationships, constraints, and governance intent which is used as single source of truth to generate all implementation schemas.

- APIs (OpenAPI),
- streams (AsyncEvent),
- database (SQL DDLs)
- Big Data (Apache Iceberg definitions)

Write once, deploy everywhere, guaranteed consistency.

This isn't just efficiency—it's how we build compounding advantages that creates competitive moats.

ODSC AI WEST 2025


Versioning
Decorators
-Policies
-Metadata
Data Validations
Documentations
Super Schema
Single Source of Truth
Governance Review, Compatibility checks, etc
Human & Governance View
Implementation View
Column-level security policies
Topic-level ACLs
OAuth scopes requirement
PII inventory
Security & Access warnings
JSON-Schema
Protobuf
OpenAPI
Docs
DDL
Avro
Java
API SDK
HTTP Client
Confluence
BigQuery
Icerberg
Zod
TypeScript

```
@doc("Customer entity - source of truth for all systems")
@businessOwner("Customer Success Team")
@retentionPolicy("7 years after account closure")
model Customer {
  @key
  @doc("Unique customer identifier")
  customer_id: int64;

  @doc("Customer's legal first name")
  @dataClassification("PII")
  @anonymization("tokenize")
  @qualityRule("non-empty, max 100 chars")
  @maxLength(100)
  first_name: string;

  @doc("Primary email address")
  @dataClassification("PII")
  @anonymization("hash")
  @qualityRule("valid email format")
  @maxLength(255)
  email: string;

  @doc("Product price with cent precision")
  @dataClassification("FINANCIAL")
  @precision(10, 2)  // ← Enforced everywhere!
  lifetime_value: decimal;
}
```

```
-- BigQuery (Auto-Generated)
CREATE TABLE customers (
  customer_id INT64 NOT NULL,
  first_name STRING(100) NOT NULL,
  email STRING(255) NOT NULL,
                        -- Precision preserved
  lifetime_value NUMERIC(10,2) NOT NULL
)
PARTITION BY DATE(created_at)
OPTIONS(labels=[("pii", "true")]);
```

```
// Avro (Auto-Generated)
{
  "name": "Customer",
  "fields": [
    {"name": "lifetime_value",
     "type": {"type": "bytes", "logicalType": "decimal",
              "precision": 10, "scale": 2}}
                //Precision preserved
  ]
}
```

ODSC AI WEST 2025

Tech & Tools

- TypeSpec - typespec.io
- **OpenAPI -** openapi.tools
- **Transform Tools :** transform.tools
- Avro
- Proto

ODSC AI WEST 2025

Benefits

- Single Source of Truth for schema - everybody refer to one place
- Much better Schema Changes planning
- Zero Schema Drift "Consistency guaranteed - can't be different"
- Instant Compliance Reports
- Reduced Production Incidents
- Faster Feature Delivery "No more waiting weeks for schema coordination"
- Developer Happiness & Executive Confidence "We know exactly where our data is, How its structured and who can access it"

ODSC AI WEST 2025

Your Data Model is Your Destiny

By Matt Brown : https://notes.mtb.xyz/p/your-data-model-is-your-destiny

HomeWork

- Search your catalog for “user” or “customer” table and see the schema / data drift for your organization.
- How many different version of tables your organization has to represent “Customers”
- How many incident occurred and could have been avoided due to schema change.
- How much time it takes for your organization to change the schema of a customer.
- …..

ODSC AI WEST 2025

Q & A

Thank You !

ODSC AI WEST 2025